

marrying and producing children who reflect the novel intermingling of unusual alleles.

Genetic Drift

In very small populations, rare alleles can become common or disappear because of genetic drift—random changes in gene frequency that are not due to selection, gene mutation, or immigration. We can explain this as follows. When flipping a coin 1,000 times, it is likely to get 50 percent heads and 50 percent tails (if it's a fair coin). But flip it only five or ten times, and it is unlikely to get exactly half heads and half tails. Chances are good that the results will be something quite different. In the same way, if 10,000 people mate and produce children, the *bb* genotype will pretty much conform to the Hardy-Weinberg equation described above (provided the other assumptions are approximately true). For example, in a sample of just twenty people, instead of getting a group of children of whom 4 percent have blue eyes, the result could end up none with blue eyes, or maybe half having blue eyes. It all depends on how the alleles happen to combine when eggs meet sperm.

Because of genetic drift, small, isolated populations often have unusual frequencies of a few alleles. Although similar to other people in most important respects, such isolated populations may harbor high frequencies of one or more alleles that are rare in most other populations. For example, in 1814, fifteen people founded a British colony on a group of small islands in the mid-Atlantic, called Tristan de Cunha. They brought with them a rare recessive allele that causes progressive blindness, and the disease, extraordinarily rare in most places, is common on Tristan de Cunha. Such “inbreeding” produces more homozygotes than usual and increases the probability of children born with genetic diseases. The Old Order Amish have a high frequency of Ellis-van Creveld syndrome, and Ashkenazi Jews were, until a few years ago, susceptible to Tay-Sachs disease. Fortunately, genetic testing has greatly reduced the incidence of Tay-Sachs and many other such genetic diseases.

Population genetics also provides information about evolution. It is known, for example, that populations that have unusual allele frequencies must have been isolated from other populations. And we can surmise that populations that share similar frequencies of certain rare alleles may have interbred at some point in the past. Human populations in sub-Saharan Africa show the greatest diversity of all human populations. On the basis, in part, of this diversity, one theory of human evolution suggests that all humans originated in Africa, and then emigrated to Asia, Europe, and the rest of the world. SEE ALSO FOUNDER EFFECT; GENETIC DRIFT; HARDY-WEINBERG EQUILIBRIUM; MOLECULAR ANTHROPOLOGY; POPULATION BOTTLENECK; SELECTION; TAY-SACHS DISEASE.

Jennie Dusbeck

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As scientific research reveals more information about treating diseases and maintaining good health, it has become increasingly important to identify diseases in their early stages in order to treat them most effectively. Thus, researchers have developed tests for some diseases to identify people at high risk for the disease before the symptoms of the disease actually appear. These tests are routinely administered to individuals in a defined population who have no apparent symptoms of the disease being screened. This process is called population screening. A primary goal of population screening is to predict with high accuracy which individuals in this group are at significant risk of developing or transmitting a disease. Once individuals at high risk for a disease are identified, confirmatory (diagnostic) tests are then performed to detect the screened-for disease with greater certainty.

Examples of routine population screening currently used in the health care field include Pap smears for women to predict their risk for cervical cancer, mammograms for women to predict their risk for breast cancer, the PPD skin test to predict exposure to tuberculosis mycobacterium (TB) in health care workers, and the prostatic antigen screening (PSA) test for men to predict their risk for prostate cancer.

Criteria for a Screening Program

Screening for Inherited Disorders

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cerns have shifted to include the growing number of recognized inherited disorders. Screening for inherited disorders began in the early 1960s, with testing for phenylketonuria (PKU), a metabolic disease that causes severe mental deficiency. Since that time, developments such as the completion of the Human Genome Project (HGP) has resulted in an increase in the number of genetic screening tests available. Types of population screening for inherited disorders can include newborn screening and carrier screening of individuals within populations known to be at high risk for certain inherited disorders.

Newborn Screening. In the United States, every state requires that certain screening tests be done on all newborn infants. Interestingly, individual states vary significantly as to which screening tests they require. For instance, while it is mandatory to screen for PKU in every state, other relatively common inherited disorders, such as medium-chain acyl-CoA dehydrogenase deficiency and congenital adrenal hyperplasia, are only screened for in selected states. Furthermore, some private organizations offer an expanded selection of testing for genetic disorders in newborns beyond what individual states mandate.

Newborn screening is the only population-based type of screening for inherited disorders. One public health benefit of population-based screening means that everyone is tested. This is especially useful for studying inherited disorders, since it permits scientists to determine with great accuracy how frequently some inherited disorders occur in the general population.

In simple terms, there are two types of testing done for newborn screening: non-DNA-based testing and DNA-based testing. With DNA-based testing, an individual's DNA is tested directly. With non-DNA-based testing, two indirect methods are used: Enzymatic or electrophoretic testing methods are used to figure out whether an individual has an inherited disorder.

The most well-known newborn screening test is for PKU. PKU is an **autosomal**, recessively inherited metabolic disorder characterized by a lack or defect of phenylalanine hydroxylase, an **enzyme** involved in the metabolism of an amino acid called phenylalanine. Because this enzyme does not function properly, persons with PKU have a buildup of phenylalanine in their bodies, which results in severe mental retardation if untreated. Treatment currently involves dietary restriction of phenylalanine. Because phenylalanine is an essential amino acid (meaning that our bodies cannot make it, and it is essential for life), people with PKU cannot eliminate the substance entirely from their diet. Instead they must take care to modulate the amount they consume, because they cannot metabolize extra phenylalanine. Therefore, persons with PKU are usually under the care of a dietitian, who helps them to eat a balanced diet with the right amount of phenylalanine. Most importantly, by following the proper diet, individuals with PKU will not develop mental retardation.

Screening for PKU is done using the Guthrie test, a bacterial inhibition assay, a non-DNA-based (enzymatic) laboratory test. Another example of a non-DNA-based test is hemoglobin electrophoresis, which is done to determine the types of hemoglobin an individual carries. This information is a direct reflection of an individual's genetic makeup. Some other examples of disorders screened for, using non-DNA-based tests, include hypothy-

autosomal describes a chromosome other than the X and Y sex-determining chromosomes

enzyme a protein that controls a reaction in a cell

roidism, congenital adrenal hyperplasia and hemoglobinopathies (such as sickle cell disease and thalassemias). In contrast, DNA-based genetic testing is a relatively new clinical tool in newborn screening. In at least one state, DNA-based newborn screening is done for cystic fibrosis.

Carrier Testing. In contrast to newborn screening, which is done on all newborns regardless of family history or ethnicity, carrier testing is aimed at a specific population that is viewed to be at high risk for a given disorder. In addition, carrier testing is always DNA-based, whereas newborn screening is typically non-DNA-based. Carrier testing is offered to determine if individuals carry a single non-working copy of a gene for a genetic disorder.

In general, individuals who carry one copy of a non-working gene will not have symptoms or signs of the disorder. When two individuals who are carriers for the same inherited disorder have a child, that child is at a 25 percent risk of inheriting a copy of the non-working gene from each parent, and will usually show symptoms and signs of the inherited disorder. Diseases that may be inherited in this way are called autosomal recessive disorders, and include sickle cell disease, Tay-Sachs disease, and cystic fibrosis.

In the late 1990s, a panel convened at the National Institutes of Health and recommended that carrier screening for cystic fibrosis be offered to a number of populations, including adults with a family history of the disease, the partners of people who had the disease, couples who were planning a pregnancy, and couples seeking prenatal testing. Newborn screening and general population screening, however, were not recommended.

Ethical Considerations

Before carrier screening can be offered to a high-risk population, the population must be educated about the disorder being screened, the basic **tenets** of carrier screening, and the potential benefits and risks of carrier screening. Screening for inherited disorders raises many complex issues. For example, screening raises a number of psychosocial issues, such as how an individual's self-esteem might be affected if he or she was found to carry a non-working gene). There are also implications for nonscreened family members if an individual is identified as a carrier. For instance, how will other family members be notified that they are also at risk for being a carrier. In addition, there is the potential for discrimination; other people may inaccurately infer that an individual who is a carrier has the inherited disorder.

Finally, and most importantly, no discussion of any type of genetic testing is complete without raising the topic of eugenics and the atrocities of the past that were associated with the abuse of genetic information. Today, each individual has the right to choose whether or not they want to know if they carry genes that predispose them to an inherited disorder. No one should be forced to learn about their carrier status, even by another family member. Moreover, the goal of any type of an inherited disorders screening program is not to eliminate genes that cause disease. Eliminating disease-causing genes would mean eliminating the human race. It is estimated that every human carries at least three to four genes that are associated with inherited diseases. Furthermore, there is a presumed constant rate of change (mutation) in human genetic material. These changes ensure variation amongst individuals. However, variation in human DNA occurs randomly, with some changes

tenets generally
accepted beliefs

leading to beneficial effects, and some to detrimental effects like an inherited disorder. Therefore, genetic disorders will continue to occur and be a part of human genetic makeup. Although general population screening and population-targeted carrier screening raise many complex issues, they will ultimately allow society to better prepare for working with these disorders. SEE ALSO CYSTIC FIBROSIS; EUGENICS; GENETIC COUNSELING; GENETIC COUNSELOR; GENETIC DISCRIMINATION; GENETIC TESTING; GENETIC TESTING: ETHICAL ISSUES; HEMOGLOBINOPATHIES; METABOLIC DISEASES; TAY-SACHS DISEASE.

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Post-translational Control

Post-translational control can be defined as the mechanisms by which protein structure can be altered after translation. Proteins are polymers of amino acids, and there are twenty different amino acids. Both the order and identity of these amino acids are important for the role that the protein plays in the cell. In some cases, the chemical identity of these amino acids is changed after translation. Alternatively, the sequence or number of the amino acids in a protein can be altered. These changes can alter the structure or function of the protein, or they can target it for destruction.

Alterations of Amino Acids

Post-translational control of protein function or structure can be accomplished by chemical alteration of an amino acid side chain or by modification of the ends of the protein backbone. While there are many diverse chemical modifications of amino acids, three common examples are phosphorylation, glycosylation, and ubiquitination.

Phosphorylation involves the addition of phosphate to an amino acid side chain, usually to the side chain hydroxyl (–OH) of serine, threonine, or tyrosine. This modification results from the action of a protein known as a **kinase** and uses **ATP** as the source of phosphate. The phosphate can be removed by another enzyme, a phosphatase. Phosphorylation can alter protein function and is relevant in cellular signaling pathways. Aberrant phosphorylation can lead to disruption of the cell cycle and the induction of cancer.

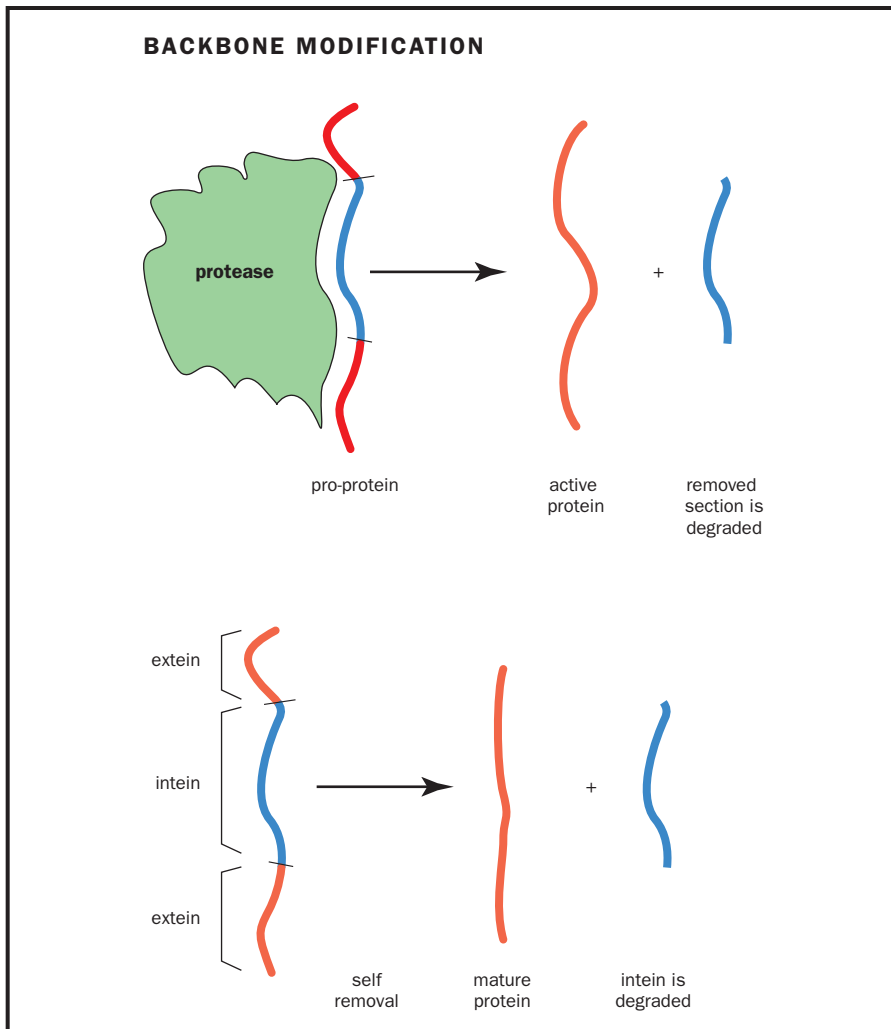
Glycosylation involves the addition of one or more sugar **monomers** to the side chains of amino acids, either during or after translation, to make a **glycoprotein**. Sugars are attached either to the side-chain nitrogen of the

kinase an enzyme that adds a phosphate group to another molecule, usually a protein

ATP adenosine triphosphate, a high-energy compound used to power cell processes

monomers "single parts"; monomers are joined to form a polymer

glycoprotein protein to which sugars are attached



Removal of some portions of the translational polypeptide may be needed to form a functional protein.

amino acid asparagine or to the hydroxyl of the amino acids serine or threonine. The structure of these carbohydrates can be complex and variable and often does not affect the function of proteins directly. However, glycosylation can affect the protein's solubility, its targeting to a particular part of the cell, its folding into a three-dimensional structure, its lifetime before it is degraded, and its interaction with other proteins.

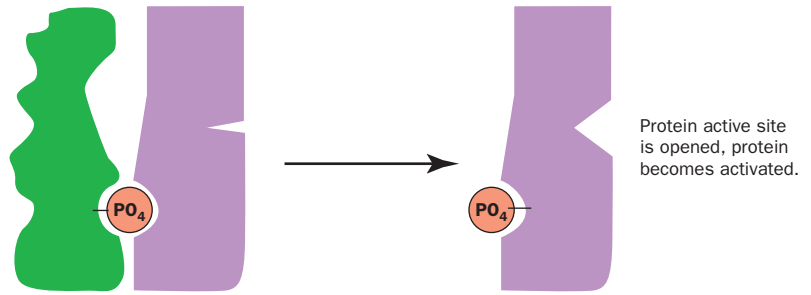
The addition of ubiquitin (a protein composed of seventy-six amino acids) to another protein can render the target protein susceptible to degradation by the 26S proteasome, which is a large protease (a protein that cleaves other proteins). Ubiquitin is ubiquitous in the cell (hence its name) and varies little between organisms as diverse as yeast and humans. It is attached via the side chain of the amino acid lysine, and often additional ubiquitin proteins are added to the first to make a chain.

Alteration of the Polypeptide Backbone

Control of protein function by post-translational modification may also occur by altering the order of the amino acids in the protein backbone. These modifications may be promoted by other proteins or they may be self-directed.

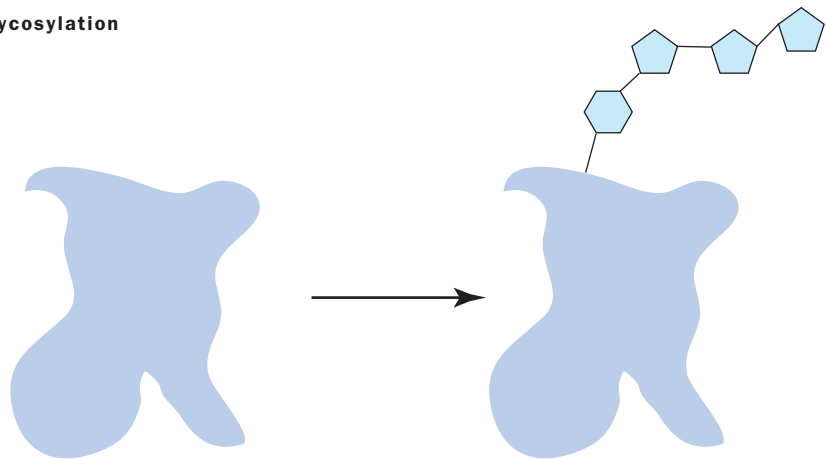
Three mechanisms of post-translational control.

Phosphorylation



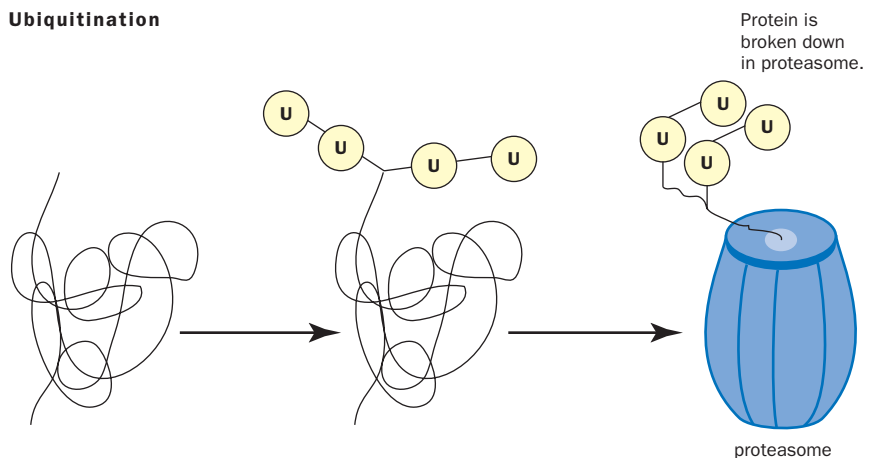
Addition of phosphates alters protein activity.

Glycosylation



Addition of sugars provides specificity to some proteins.

Ubiquitination



Ubiquitous tag misshapes proteins for breakdown.

Inteins

Inteins promote their own excision from their target protein as well as the ligation of the flanking protein segments, which are called exteins. It is possible that some inteins play a role in regulating gene expression, but it is also possible that they are vestiges of ancient control mechanisms or simply molecular parasites.

While it is unclear if protein splicing has a regulatory role, a class of proteins evolutionarily related to inteins, the hedgehog proteins, are involved in embryonic patterning and segmentation. Hedgehog proteins promote their own internal cleavage, coupled to the addition of cholesterol to the C-terminal end of the N-terminal fragment, which is probably important for anchoring the hedgehog protein in the cell membrane. SEE ALSO RNA PROCESSING; SIGNAL TRANSDUCTION; TRANSCRIPTION.

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Progeria *See Accelerated Aging: Progeria*

Prenatal Diagnosis

The future health of a new individual can be predicted, to an extent, from clues that are apparent before birth. Prenatal diagnosis is the identification of a medical condition in a developing embryo or fetus. Prenatal testing can sample fetal cells to examine DNA sequences that correspond to specific disease-causing genes or chromosomes (the structures that carry the genes). Biochemicals obtained with the fetal cell samples can also hold clues to health. Other prenatal tests analyze a pregnant woman's blood serum for telltale biochemicals that indicate the fetus faces an elevated risk of a particular condition. Ultrasound scans provide views of many aspects of fetal anatomy. Preimplantation genetic diagnosis is a technique that is actually pre-prenatal. It provides a health check on very early embryos grown in a laboratory dish, enabling parents to select those that are most likely to develop into healthy infants.

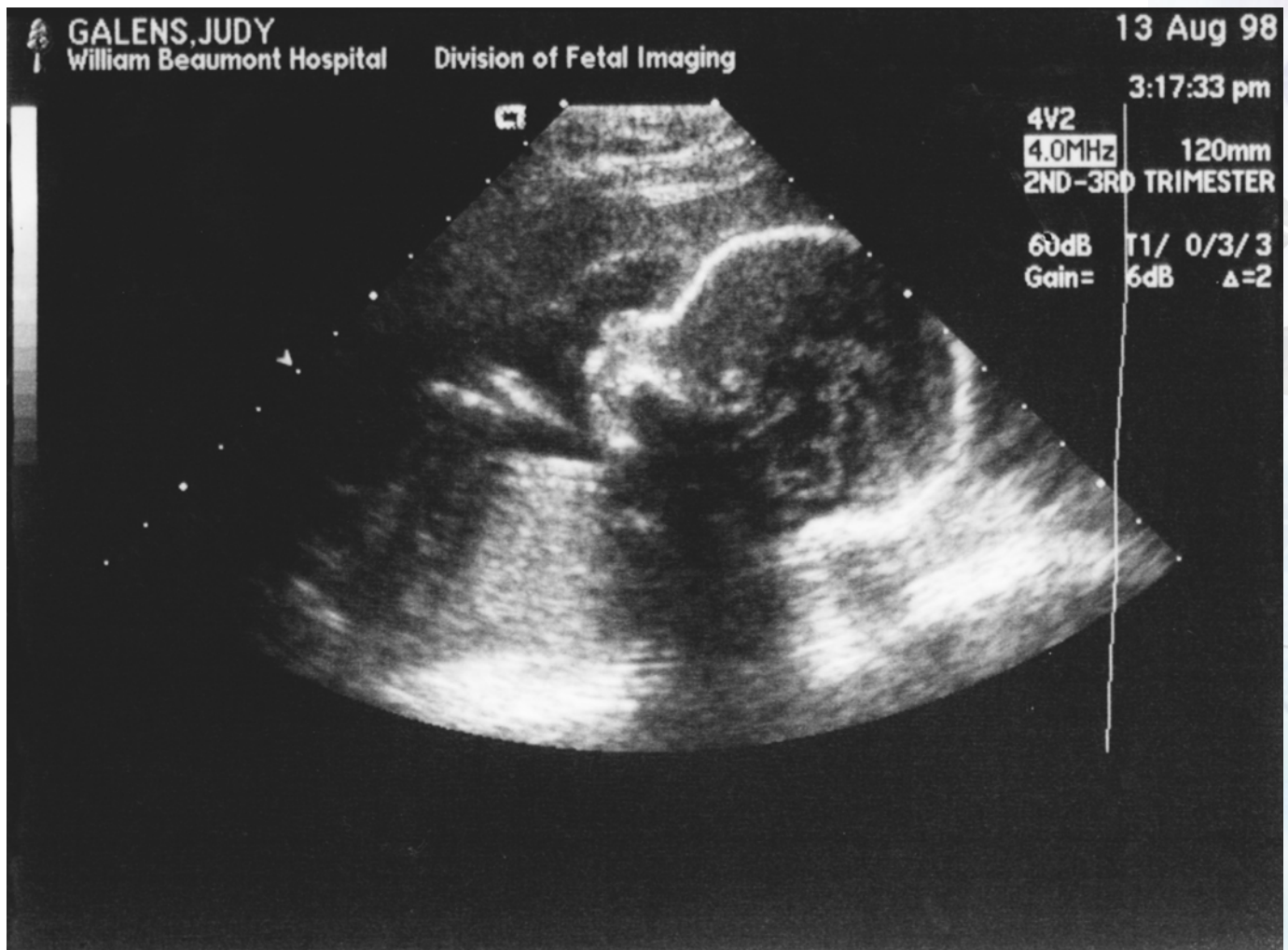
Because prenatal tests that sample fetal cells are invasive, they carry a risk of the test causing miscarriage. Therefore, these procedures are typically offered only to those pregnant women whose risk of carrying a fetus with a detectable condition is greater than the risk of miscarriage. Reasons include already having had a child or family history with a detectable genetic or chromosomal condition, or "advanced maternal age." After age thirty-five, a woman's risk of carrying a fetus with an extra or missing chromosome exceeds the risk that the procedure will cause miscarriage.

Viewing Chromosomes

Biologists first tried to visualize the chromosomes in a human cell in the late nineteenth century, with estimates of the total number ranging from 30 to 80. As methods to untangle and stain chromosomes improved, the count narrowed to 46 or 48, and by 1956 was confirmed as 46, or 23 pairs. By 1959, the first chromosomal abnormalities were identified using size and crude staining patterns to distinguish the chromosomes. In the 1970s, vastly improved staining techniques enabled **cytogeneticists** to much more easily distinguish chromosomes, and they began amassing databases of specific chromosomal abnormalities and the clinical syndromes that they cause.

Also in the 1970s, general staining began to be replaced with *in situ* hybridization, an approach that links a radioactive molecule to a short sequence of DNA called a DNA probe, chosen to match a known gene of interest. When the DNA probe binds to its complementary sequence among a sample of chromosomes spread against a piece of photographic film, the radioactivity exposes the film exactly where the probed DNA sequence resides. In the 1990s, fluorescent molecules replaced the radioactive tags, and a procedure called fluorescence *in situ* hybridization (FISH) was born. A flash of light matches probe to chromosome. Today, FISH can use combinations of fluorescent labels and computer analysis to individually label

cytogeneticists scientists who study chromosome structure and behavior



each chromosome. The technique is called chromosome painting or spectral karyotyping. A karyotype is a picture of a person's chromosomes displayed in size-ordered pairs. FISH can be used to highlight chromosomes obtained by amniocentesis, CVS, or fetal cell sorting, described next.

Viewing fetal chromosomes requires obtaining cells from the fetus. The most common procedure is amniocentesis, first successfully performed in 1966. In amniocentesis, a needle is used to remove a sample of the amniotic fluid that surrounds the fetus. This is usually done after the fifteenth week of pregnancy. The fluid sample contains skin cells that the fetus has shed, and these are analyzed for their chromosomal content. Results from amniocentesis typically are available within two weeks. FISH is not routinely offered, but in the labs that do offer it, some preliminary information may be available more quickly than is possible with other testing procedures.

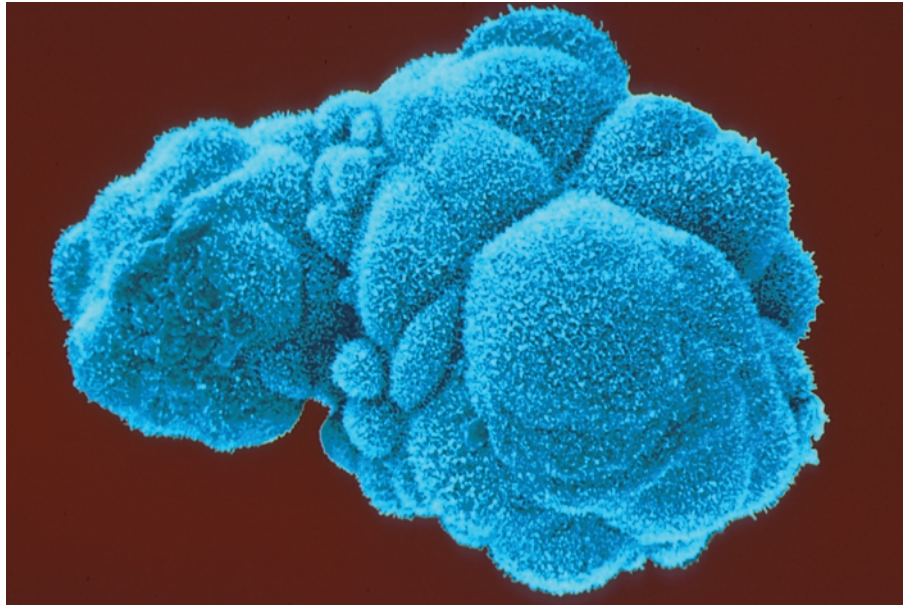
Aberrations of chromosomes 13, 18, 21, X, and Y are seen most commonly. This is not necessarily because they are affected more often, but because problems in other **autosomes** are so severe that development ceases long before prenatal testing can be done.

Biochemicals in the amniotic fluid can also be analyzed for signs of metabolic disorders, though this procedure is not commonly performed unless

Ultrasound monitors fetal development late in the pregnancy. This image was taken late in the second trimester.

autosomes chromosomes that are not sex-determining (not X or Y)

A human embryo at the blastula stage (day 6). This embryo could have been tested earlier in its formation (by in vitro fertilization) for various birth defects before being implanted into the female to develop.



there is already a suspicion that one may be present. Chemical markers may also be sought for neural tube defects (NTDs), which are abnormalities in brain or fetal spinal cord development. Risk of amniocentesis causing miscarriage is about 1 in 200.

Chorionic villus sampling (CVS) can be performed earlier than FISH, from the tenth to twelfth week of pregnancy. A physician removes a small sample of the chorionic villi, reached either through the vagina or the abdominal wall. The chorionic villi are fingerlike projections of cells that form part of the placenta, which provides nutrients to the developing fetus. Because the chorionic villi originate from the fertilized ovum, their chromosomes and genes should be the same as those in fetal cells. However, in practice, sometimes a mutation affects only the chorionic villi, leading to a false positive test result, or only the fetus, leading to a false negative result. Maternal cells may also contaminate the sample. Because of these uncertainties, follow-up testing such as amniocentesis is required for clarification.

CVS has been linked to a fatal limb defect, and carries a risk of miscarriage of about 1 percent. It is typically recommended for women over the age of thirty-five, for those who have already had a child with a detectable genetic or chromosomal defect, if there is a family history of a genetic disorder, or when abnormalities are detected by ultrasound. For example, CVS is often used if there is a family history of Duchenne muscular dystrophy or Tay-Sachs disease. Unlike amniocentesis, CVS cannot detect NTDs because it does not sample biochemicals. Its advantage is that it can be performed earlier in the pregnancy.

A third technique, called fetal cell sorting, is being studied and may eventually replace amniocentesis and CVS in obtaining fetal cells. This approach isolates the rare fetal cells that enter the mother's blood stream and analyzes them for gene and chromosome abnormalities. A device called a fluorescence-activated cell sorter detects and isolates fetal cells by their different surface features compared to cells from the pregnant woman.

Because fetal cell sorting requires only a blood sample from the pregnant woman, it cannot endanger the fetus.

Less Invasive Methods

An ultrasound scan bounces soundwaves off of the fetus to create an image. A scan is often performed after the sixteenth week of pregnancy, and the anatomy and size of the fetus is measured to see if it is growing and developing normally. The scan can often detect major structural problems, such as a malformed heart or spine. An unusual finding on an ultrasound scan can be a warning to investigate further. However, not all birth defects can be detected by ultrasound.

An ultrasound is sometimes done at weeks five or six to confirm that a pregnancy is present. This early, the embryo looks like a lima bean with a pulsating blip in the middle, which is the beating heart. Ultrasound performed late in pregnancy can provide clues to the approaching birth date. New three-dimensional ultrasound scans offer spectacular views of the fetus.

Another noninvasive method to detect fetuses at risk for some birth defects is maternal serum marker screening. A sample of blood from a pregnant woman taken at approximately weeks 15 to 18 is analyzed for the amount of several substances, including alpha fetoprotein (AFP); a form of estrogen called unconjugated estriol; and human chorionic gonadotropin (hCG), a hormone produced only during pregnancy.

Maternal serum screening began in the 1970s with the AFP test, invented by a man whose son was born with a neural tube defect. High levels of AFP in a woman's blood indicate an increased risk for a neural tube defect in the fetus. The neural tube forms by approximately day 28 of gestation, when a portion of the flat embryo (the neural plate) folds to form a tube that will develop into the brain and spinal cord. The tube normally closes up like a zipper starting at several points along its length. If a hole remains, the brain and spinal cord underneath are exposed, causing damage.

Several years after the AFP test was developed for neural tube defects, researchers noted that low AFP correlates to an increased risk that a fetus will have an extra chromosome, particularly at positions 18 or 21. This condition is called a trisomy. Trisomy 21, an extra chromosome 21, is the most common cause of Down syndrome. Over the years, analysis of other substances have been added to refine this test, which is now offered routinely to pregnant women. Abnormal results on maternal serum screening tests indicate that amniocentesis should be done to diagnose a neural tube or chromosome defect, and that genetic counseling should be offered.

Preimplantation Genetic Diagnosis

Amniocentesis, CVS, and maternal serum screening are performed after a pregnancy is confirmed or in progress. In contrast, preimplantation genetic diagnosis (PGD) occurs before the embryo implants in the womb. This technique is performed on an embryo that has been derived from in vitro fertilization (IVF) and is growing in a laboratory dish. At about the 8-cell (day 3) stage, a cell is removed and the DNA and chromosomes are checked using FISH or a probe for a specific gene. If the cell is free of the defects being probed, the remaining 7-celled embryo is implanted into the woman, where it continues development.



A doctor at Baltimore University Hospital has inserted the needle that will extract amniotic fluid from this patient's placenta. The fluid obtained from the amniocentesis will be analyzed to detect genetic disorders.



The first PGD was done in 1989, when it was used to enable families with X-linked disorders to select a girl, who would not be affected by the condition. Then it was used to conceive Chloe O'Brien, a youngster free of the cystic fibrosis that affected her brother. PGD attracted widespread public attention in 2001, when a Minnesota couple, Lisa and Jack Nash, conceived their son Adam so that his umbilical cord stem cells could be used to cure his sister Molly's Fanconi anemia. Adam not only had not inherited Fanconi anemia, but he was also a tissue match for Molly, saving her life.

PGD has been used to eliminate embryos with a variety of single-gene disorders, including metabolic disorders, dwarfism, cystic fibrosis, hemophilia, muscular dystrophies, and several other genetically inheritable diseases. The technique is being increasingly used in couples for whom IVF has repeatedly failed because they manufacture eggs or sperm that have abnormal numbers of one or more chromosomes. PGD enables physicians to sort through embryos to identify and transfer those few that have normal chromosomes. PGD has about a 66 percent success rate for identification of genetic disorders.

Genetic Counseling and the Ethics of Prenatal Diagnosis

A genetic counselor helps educate individuals, couples, and families about prenatal tests, and helps them to understand and cope with the results. The counselor also informs the prospective parents of the limitations of the tests, explaining that they can rule out certain conditions but cannot guarantee a healthy baby.

Ethical issues can arise in the decision to undergo prenatal testing. For example, the Nash family received criticism for their decision to intentionally conceive one child to save another. Some people also question the use of prenatal tests or PGD to reject embryos because of a gene that causes an adult-onset disease, such as Alzheimer's disease. In a more general sense, picking and choosing offspring based on genes can be considered eugenic,

with the caveat that the intent is not to improve the gene pool, but to prevent suffering. This may mean terminating a pregnancy in which the fetus has a very bleak prognosis, which people opposed to abortion might find unethical. Opponents to this view point out that “letting nature take its course” can be painful for the fetus and may endanger the life of the woman.

The ethics of prenatal diagnosis becomes more complicated when the goal is not to prevent suffering, but to choose a child of a particular sex. Doctors have long reported patients using CVS or amniocentesis to learn the sex of the fetus, then terminating the pregnancy if the outcome is not what is desired. PGD is sometimes used for this purpose, too. Some people have compared this practice to a high-tech version of the ancient practice of leaving girl babies outside city walls to perish. The American Society for Reproductive Medicine endorses the use of PGD for sex selection to avoid passing on an X-linked disease, but discourages use for family planning as “inappropriate use and allocation of medical resources.”

The ethical concerns that arose with the ability to foretell the sex of a child are certain to mushroom as data from the Human Genome Project continue to lengthen the list of disorders that can be detected before birth. Physicians and parents-to-be in the future will have to decide just how much they want to know about their offspring and how they will use that information. SEE ALSO CHROMOSOMAL ABERRATIONS; DOWN SYNDROME; EUGENICS; GENETIC COUNSELING; GENETIC COUNSELOR; GENETIC DISEASE; GENETIC TESTING; *IN SITU* HYBRIDIZATION; MOSAICISM; REPRODUCTIVE TECHNOLOGY: ETHICAL ISSUES; TAY-SACHS DISEASE.

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Prion

In 1997 Stanley Prusiner was awarded the Nobel Prize in physiology or medicine for a revolutionary theory about the mechanisms of infection. His theory, the “prion hypothesis,” concerns an unusual protein, the prion, which occurs in the complete absence of DNA and RNA. According to Prusiner’s theory, the prion differs from other well-known infections agents including bacteria and viruses. While the latter rely on **nucleic acid** for survival and replication, the prion is made of a protein and lacks nucleic acid. Both the existence of the prion and the underlying mode of infection are unprecedented in medical sciences. While several critical issues remain to be addressed, the prion hypothesis may furnish a plausible framework to understand the **pathogenesis** of several deadly brain diseases of the central nervous system.

nucleic acid DNA or RNA

pathogenesis pathway leading to disease



Stanley Prusiner won the 1997 Nobel prize in physiology or medicine for his research on prions.

scrapie prion disease of sheep and goats

glycoprotein protein to which sugars are attached

polypeptide chain of amino acids

enzymes proteins that control a reaction in a cell

aggregate stick together

A New Infectious Agent

Prion is an acronym for “proteinaceous infectious particle,” a term coined by Prusiner in the early 1980s to describe the nature of the agent causing the fatal brain disorders known as transmissible spongiform encephalopathies (TSE), also called prion diseases. Well-known examples of prion diseases include scrapie in sheep and goats, bovine spongiform encephalopathy (BSE, or “mad cow” disease) in cattle, and Creutzfeldt-Jakob disease (CJD) in humans. Prion diseases are infectious and can also be transmitted to healthy animals by inoculating them with extracts of diseased brain.

In the mid-1960s, Tikvah Alper and colleagues reported that nucleic acid was unlikely to be a component of the infectious agent that causes **scrapie**. In 1967 J. S. Griffith speculated that the scrapie agent might be a protein capable of “self replication” without nucleic acid. However, Prusiner was the first, in the early 1980s, to successfully purify the infectious agent and to show that it consisted mostly of protein (technically speaking it is a **glycoprotein**, because it has a sugar group attached). He chose to name the new agent “prion” to distinguish it from viruses or viroids.

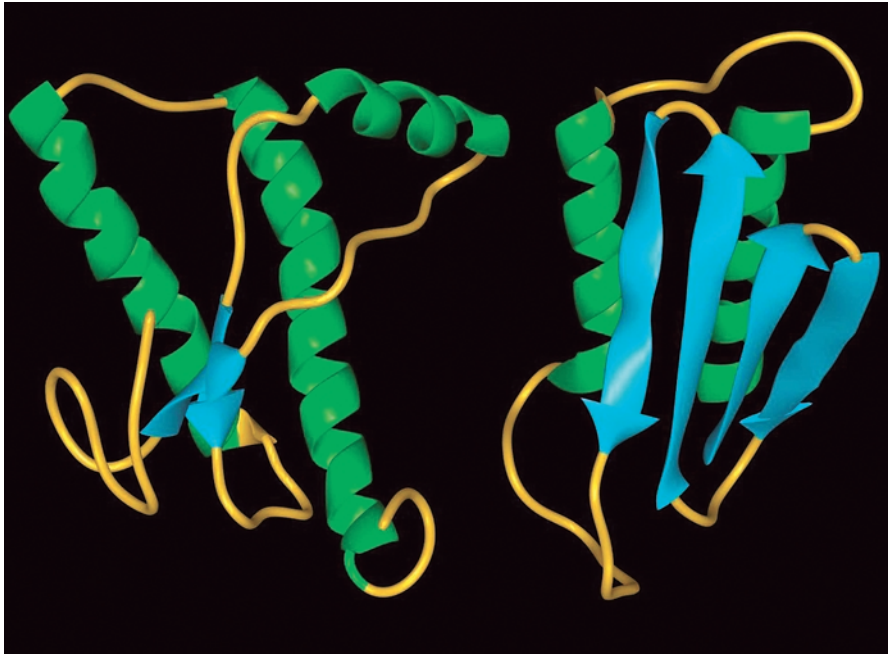
The essential protein component of prion was later identified in 1984 as prion protein (PrP), which is encoded by a chromosome gene in the host genome. Researchers concluded that the prion is a new infectious agent that consists mostly of PrP. This view is often referred to as the “protein only” or prion hypothesis. Some scientists find this notion hard to accept and have argued that nucleic acid is needed to carry information necessary for infection. However, no one has been able to demonstrate that either DNA or RNA play a direct role in prion replication.

In 1992 Charles Weissmann and colleagues obtained conclusive evidence for the central role of PrP in the transmission of prion diseases, when they created transgenic mice devoid of the PrP gene. These so-called PrP knock-out mice were found to be completely resistant to infection when inoculated with scrapie brain preparations. When the PrP gene was reintroduced into the knockout mice, they once again became susceptible to prion infection.

Role of Protein Conformation

How can a protein such as PrP made by a cellular gene become an infectious agent? Prusiner and associates had found that PrP could exist in two forms, a normal or cellular form (PrP^C) normally expressed at low levels in neurons and other cell types, and an abnormal or scrapie form (PrP^{Sc}) built up in diseased brain. PrP^C is a cell-surface glycoprotein, the function of which has yet to be established. PrP^C consists of a single **polypeptide** chain folded into predominantly spiral conformations known as α -helices. These structures give rise to a globular shape that is soluble and can be cleared from the cell by degrading **enzymes** called proteases.

In contrast, PrP^{Sc} that has been isolated from diseased brain is rich in an alternative conformation that resembles extended strands. These structures are known as β -sheets. The β -sheet rich PrP^{Sc} tends to **aggregate** and is resistant to heat and degradation by proteases. It is assumed that PrP^{Sc} can initiate the infection process by binding to predominantly α -helical PrP^C and converting it into more stable PrP^{Sc} with β -sheet conformation. This will set off a chain reaction leading to accumulation of large amounts of



The illustration on the left is a computer-generated image of a healthy human prion protein. That on the right is a model of the disease-causing prion protein. The blue sections are β -strands, the green are α -helices, and the yellow are the chains connecting these regions.

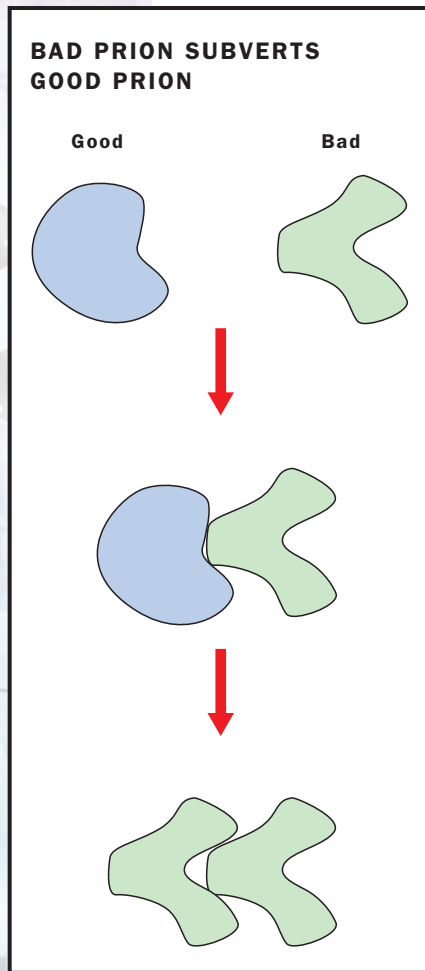
PrP^{Sc} to levels that result in brain tissue damage. The conformational conversion from α -helices to β -sheets transforms the benign PrP^{C} into disease-causing PrP^{Sc} . This model of conformational conversion provides useful insights into the pathogenesis of prion diseases.

Prion Diseases

Historically, prion diseases have been given distinct names. Scrapie is a naturally occurring prion disease of sheep and goats that was first documented in Iceland during the eighteenth century. BSE or mad cow disease is a prion disease of cattle and is believed to be acquired through scrapie-contaminated foodstuffs. Kuru, a prion disease found among the Fore tribe of New Guinea, was shown by D. Carleton Gajdusek to be transmitted by the consumption of human tissue, particularly brain tissue, during funerary rituals. Gajdusek was awarded the 1976 Nobel Prize in physiology or medicine for this contribution. The early symptom of Kuru is a loss of coordination, followed by mental confusion and, ultimately, death. It has virtually disappeared since 1958, when the practice of eating human tissue was more or less eradicated in New Guinea.

CJD is the most common human prion disease, affecting about one in a million people. The main symptom is **dementia**, along with other neurological signs. There are three forms of CJD. Sporadic CJD, the cause of which has yet to be found, is a spontaneous disease that accounts for a majority of CJD cases. Familial CJD affects people who carry a mutation in the PrP gene on chromosome 20. The third form, called iatrogenic CJD, is the result of accidental transmission during medical treatments. A newly emerged CJD phenotype, commonly called variant CJD, has occurred in the United Kingdom since 1985. Variant CJD has a unique disease profile, and may result from the consumption of BSE-contaminated meat products. It has been diagnosed mostly in young people who initially seek treatment for psychiatric symptoms. Gertsman-Sträussler-Scheinker (GSS) syndrome

dementia neurological illness characterized by impaired thought or awareness



A schematic model for the action of prions, showing how the “bad” prion alters the shape of the “good” prion.

phenotypes observable characteristics of organisms

is a familial prion disease resulting from a mutation in the PrP gene. The main symptom of GSS is the loss of coordination and dementia. Fatal familial insomnia (FFI) is another a familial prion disease in which fatal dementia follows the loss of physiological sleep.

Although human prion diseases manifest as three etiologically different forms—spontaneously (sporadic CJD), through inheritance (familial CJD, GSS, and FFI), and by infection (iatrogenic CJD, kuru, and possibly the new variant CJD), they nonetheless share a common pathogenetic event. Within the framework of “protein only” hypothesis, they all involve the protein conformational change that converts PrP^C to pathogenic PrP^{Sc}. Such a structural change in PrP may be triggered by a rare spontaneous event leading to a sporadic disease, a mutation that causes a familial disease, or exposure to foreign PrP^{Sc}, leading to an acquired disease. The “protein only” hypothesis provides a plausible mechanism underlying the pathogenesis of all forms of prion diseases. Moreover, it also helps explain the tremendous variability in prion-associated disease **phenotypes**. Structurally distinct variants of PrP^{Sc} may accumulate in different regions of the brain and initiate pathogenic changes that may eventually lead to distinct pathology in different areas of the brain, and subsequently the particular disease symptoms.

The concept of the prion and the role of protein conformation in disease pathogenesis have renewed inquiry into the causes of other and more common neurodegenerative disorders, such as Alzheimer’s disease, Huntington’s disease, and Parkinson’s disease. A common hallmark of all these diseases, as in prion diseases, is the conversion of an otherwise soluble and functional neuronal protein into a β -sheet rich and protease-resistant protein that has a higher tendency to aggregate and is harmful to the brain. These common pathogenetic features raise the hope that therapeutic interventions based on the same principles may be effective in all these diseases.

SEE ALSO PROTEIN.

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Privacy

As more diagnostic, screening, and monitoring tests based on genetic data become available, privacy issues are becoming increasingly important. There are concerns that the results of genetic tests showing a person to be predisposed to a particular disease will fall into the hands of commercial medical suppliers or financial, legal, insurance, or government agencies, all of which control important products or services.

The confidentiality of medical information is of paramount importance to most consumers and patients. However, maintaining confidentiality is



made difficult by the use of large medical record databases and other electronically stored records, to which any number of individuals may have access.

The Potential for Misuse of Medical Records

Medical records can be misused unless they are coded to hide patients' identifying information. If a patient has been treated for a particular disease and his or her medical records are not held in confidence, a company selling products related to the disease could directly contact the patient. Although this may not pose a problem in most cases, in some situations, such as if a patient was treated for a sexually transmitted disease, the patient might not want family members or others with access to his or her mail to know about the treatment. There are also concerns about the potential for discrimination arising from the use of these data in determining a patient's eligibility for employment, housing, or other services.

In the United States, legislation has been passed to deal with issues surrounding genetic and health information. The Health Insurance Portability

Important personal information, such as a person's DNA, can be gleaned from blood samples. Proper labeling and secure storage of blood samples helps to protect patient privacy.

and Accountability Act of 1996 was enacted to address privacy issues related to personal health information. This act requires that health care providers, health plans, and health care clearinghouses implement certain privacy standards regarding health information.

Although the act protects “all medical records and other individually identifiable health information,” there is some concern that it does not provide sufficient protection for the privacy of genetic information. In 2001 additional protection was proposed in at least three bills in the U.S. Congress. These bills were intended to prohibit discrimination on the basis of genetic information with respect to health insurance. The area of privacy and genetic information continues to develop, with additional legislation on the federal and state levels certain to arise.

European countries have addressed issues of privacy and personal information in a Directive on Data Protection. This directive, which became effective in October 1998, established a comprehensive legal regime in the European Union that governs the collection and use of personal information.

Privacy questions abound when it comes to genetic testing to determine if a person carries particular genes. One concern is that patients affected by genetic diseases, as well as those potentially at risk of disease, could be discriminated against. Another is that genetic information could also lead to discrimination against the children of those directly affected by a genetic disease.

Genetic Information in the Justice System

Many similar concerns arise in the context of criminal law, including the potential uses of DNA databases. There are issues relating to the collection and maintenance of DNA samples or information from everyone who is arrested, whether or not they are convicted. There are issues relating to the collection and maintenance of DNA samples and/or information collected from individuals upon arrest. For example, the DNA and/or information obtained from certain individuals may be saved, even if the person is not convicted. Indeed, prosecutors have issued many arrest warrants in old cases based solely on stored DNA data. These warrants have resulted in successful prosecutions, but the question being asked in the courts is whether it is legal to base arrests solely on “cold hit identification” using DNA evidence.

In contrast to medically oriented genetic tests, the DNA tests used in criminal law generally do not test for the presence or absence of a particular gene, since the noncoding regions of a person’s DNA can be distinguished much more easily from the DNA of other individuals. Different individuals have different DNA sequences in these noncoding regions because there is no evolutionary penalty for mutations in such regions, as they are not used to produce proteins.

This helps provide the high level of discrimination required in criminal cases, enabling a jury to say that, based in part on the DNA evidence, an accused person is guilty beyond a reasonable doubt. An important caveat however, and one not always understood by prosecutors or juries, concerns what a DNA match actually proves. While nonmatching DNA proves inno-

cence, matching DNA does not prove guilt. In any large city, there will be at least a handful of people with similar DNA profiles. Even if DNA is found to be matching, a conviction must rely on other evidence, such as other physical evidence or eyewitness testimony.

Although the use of DNA data can assist investigations, there is an element of “big brother is watching” in its use. There are also concerns that by instituting wide programs of DNA collection based on arrests, not necessarily convictions, the practice will expand to other areas. For example, providing a DNA sample could be required, at some point, for obtaining a driver’s license, marriage license, or social security number. There is also a question of what entities, including police departments, governmental agencies, employers, financial institutions, credit reporting businesses, and insurance carriers, would have access to the data. There is concern that by having genetic information recorded in a criminal record database, citizens would be subject to a wide variety of discrimination. SEE ALSO DISEASE, GENETICS OF; DNA PROFILING; GENETIC DISCRIMINATION; GENETIC TESTING; GENETIC TESTING: ETHICAL ISSUES; HUMAN DISEASE GENES, IDENTIFICATION OF; LEGAL ISSUES.

Kamrin T. MacKnight

Probability

Probability measures the likelihood that something specific will occur. For example, a tossed coin has an equal chance, or probability, of landing with one side up (“heads”) or the other (“tails”). If you drive without a seat belt, your probability of being injured in an accident is much higher than if you buckle up. Probability uses numbers to explain chance.

If something is absolutely going to happen, its probability of occurring is 1, or 100 percent. If something absolutely will not happen, its probability of occurring is 0, or 0 percent.

Probability is used as a tool in many areas of genetics. A clinical geneticist uses probability to determine the likelihood that a couple will have a baby with a specific genetic disease. A statistical geneticist uses probability to learn whether a disease is more common in one population than in another. A computational biologist uses probability to learn how a gene causes a disease.

The Clinical Geneticist and the Punnett Square

A Punnett square uses probability to explain what sorts of children two parents might have. Suppose a couple knows that cystic fibrosis, a debilitating respiratory disease, tends to run in the man’s family. The couple would like to know how likely it is that they would pass on the disease to their children.

A clinical geneticist can use a Punnett square to help answer the couple’s question. The clinical geneticist might start by explaining how the disease is inherited: Because cystic fibrosis is a **recessive** disease caused by a single gene, only children who inherit the disease-causing form of the cystic fibrosis gene from both parents display symptoms. On the other hand, because the cystic fibrosis gene is a recessive gene, a child who inherits only

recessive requiring the presence of two alleles to control the phenotype

Both parents are carriers of the recessive CF gene (f) in this Punnett square. The probability of having a child with CF, if both parents are carriers, is 25 percent.

PUNNETT SQUARE FOR THE INHERITANCE OF CYSTIC FIBROSIS

		father	
		F	f
mother	F	FF	Ff
	f	Ff	ff

Summary: 1 FF child = 1 of the 4 children = $1/4 = 25\%$
 2 Ff children = 2 of the 4 children = $2/4 = 50\%$
 1 ff child = 1 of the 4 children = $1/4 = 25\%$

The Bottom Line: normal children = FF and Ff = $25\% + 50\% = 75\%$
 children with CF = ff = 25%

one copy of a defective gene, along with one normal version, will not have the disease.

Suppose the recessive, disease-causing form of the gene is referred to as “f” and the normal form of the gene is referred to as “F.” Only individuals with two disease-causing genes, ff, would have the disease. Individuals with either two normal copies of the gene (FF) or one normal copy and one mutated copy (Ff) would be healthy.

If the clinical geneticist tests the parents and finds that each carries one copy of the cystic fibrosis gene, f, and one copy of the normal gene, F, what would be the probability that a baby of theirs would be born with the cystic fibrosis disease? To answer this question, we can use the Punnett square shown in the figure above. A Punnett square assumes that there is an equal probability that the parent will pass on either of its two gene forms (“alleles”) to each child.

The parents’ genes are represented along the edges of the square. A child inherits one gene from its mother and one from its father. The combinations of genes that the child of two Ff parents could inherit are represented by the boxes inside the square.

Of the four combinations possible, three involve the child’s inheriting at least one copy of the **dominant**, healthy gene. In three of the four combinations, therefore, the child would not have cystic fibrosis. In only one of the four combinations would the child inherit the **recessive** allele from both parents. In that case, the child would have the disease. Based on the Punnett square, the counselor can tell the parents that there is a 25 percent probability, or a one-in-four chance, that their baby will have cystic fibrosis.

The Statistical Geneticist and the Chi-Square Test

Researchers often want to know whether one particular gene occurs in a population more or less frequently than another. This may help them determine, for example, whether the gene in question causes a particular disease. For a dominant gene, such as the one that causes Huntington’s disease, the frequency of the disease can be used to determine the frequency of the gene, since everyone who has the gene will eventually develop the disease. How-

dominant controlling the phenotype when one allele is present

recessive requiring the presence of two alleles to control the phenotype

ever, it would be practically impossible to find every case of Huntington's disease, because it would require knowing the medical condition of every person in a population. Instead, genetic researchers sample a small subset of the population that they believe is representative of the whole. (The same technique is used in political polling.)

Whenever a sample is used, the possibility exists that it is unrepresentative, generating misleading data. Statisticians have a variety of methods to minimize sampling error, including sampling at random and using large samples. But sampling errors cannot be eliminated entirely, so data from the sample must be reported not just as a single number but with a range that conveys the precision and possible error of the data. Instead of saying the prevalence of Huntington's disease in a population is 10 per 100,000 people, a researcher would say the prevalence is 7.8–12.1 per 100,000 people.

The potential for errors in sampling also means that statistical tests must be conducted to determine if two numbers are close enough to be considered the same. When we take two samples, even if they are both from exactly the same population, there will always be slight differences in the samples that will make the results differ.

A researcher might want to determine if the prevalence of Huntington's disease is the same in the United States as it is in Japan, for example. The population samples might indicate that the prevalences, ignoring ranges, are 10 per 100,000 in the United States and 11 per 100,000 in Japan. Are these numbers close enough to be considered the same? This is where the Chi-square test is useful.

First we state the “null hypothesis,” which is that the two prevalences are the same and that the difference in the numbers is due to sampling error alone. Then we use the Chi-square test, which is a mathematical formula, to test the hypothesis.

The test generates a measure of probability, called a *p* value, that can range from 0 percent to 100 percent. If the *p* value is close to 100 percent, the difference in the two numbers is almost certainly due to sampling error alone. The lower the *p* value, the less likely the difference is due solely to chance.

Scientists have agreed to use a cutoff value of 5 percent for most purposes. If the *p* value is less than 5 percent, the two numbers are said to be significantly different, the null hypothesis is rejected, and some other cause for the difference must be sought besides sampling error. There are many statistical tests and measures of significance in addition to the Chi-square test. Each is adapted for special circumstances.

Another application of the Chi-square test in genetics is to test whether a particular genotype is more or less common in a population than would be expected. The expected frequencies can be calculated from population data and the Hardy-Weinberg Equilibrium formula. These expected frequencies can then be compared to observed frequencies, and a *p* value can be calculated. A significant difference between observed and expected frequencies would indicate that some factor, such as natural selection or migration, is at work in the population, acting on allele frequencies. Population geneticists use this information to plan further studies to find these factors.



The Computational Biologist and BLAST

Genetic counseling lets potential parents make an informed decision before they decide to have a child. Geneticists, however, would like to be able to take this one step further: They would like to be able to cure genetic diseases. To be able to do so, scientists must first understand how a disease-causing gene results in illness. Computational biologists created a computer program called BLAST to help with this task.

To use BLAST, a researcher must know the DNA sequence of the disease-causing gene or the protein sequence that the gene encodes. BLAST compares DNA or protein sequences. The program can be used to search many previously studied sequences to see if there are any that are similar to a newly found sequence. BLAST measures the strength of a match between two sequences with a *p* value. The smaller the *p* value, the lower the probability that the similarity is due to chance alone.

If two sequences are alike, their functions may also be alike. For BLAST to be most useful to a researcher, there would be a gene that has already been entered in the library that resembles the disease-causing gene, and some information would be known about the function of the previously entered gene. This would help the researcher begin to hypothesize how the disease-causing gene results in illness. SEE ALSO BIOINFORMATICS; CLINICAL GENETICIST; COMPUTATIONAL BIOLOGIST; CYSTIC FIBROSIS; HARDY-WEINBERG EQUILIBRIUM; HOMOLGY; INTERNET; MENDELIAN GENETICS; METABOLIC DISEASE; STATISTICAL GENETICIST.

Rebecca S. Pearlman

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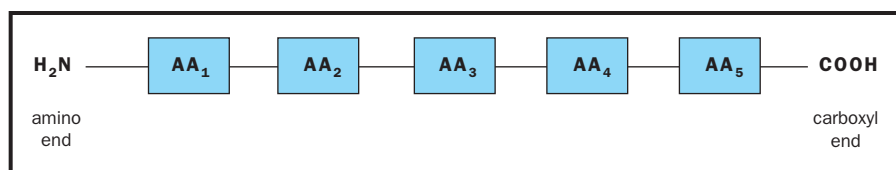
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Internet Resources

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- The National Center for Biotechnology Information. <<http://www.ncbi.nlm.nih.gov>>.

Protein Sequencing

The molecules that give cells and entire organisms their shape as well as their ability to move, grow, and reproduce are the proteins. Although they come in an almost infinite variety of shapes and sizes, they have all been designed by the process of evolution to serve a defined and useful function in the processes of life. Some proteins, like actin and collagen, help to give a cell its physical shape. Other proteins, like lactase and pepsin, help in the digestion of food. Others transport signals between cells, help us fight off



Schematic diagram of protein primary structure. Amino acids are linked head to tail, so that at one end there is a free amino group, and at the other a free carboxyl group. Proteins are typically 50–500 amino acids in length.

disease, or repair damaged DNA. For almost every job in a cell, there is a protein designed to do it.

The Building Blocks of Proteins

The building blocks of proteins are amino acids. There are twenty different amino acids used by living cells to build proteins. They are linked together in a long, linear chain during the process of translation, which is carried out by the ribosomes inside cells. Proteins begin to take on their characteristic three-dimensional shape even while they are being made, folding and twisting as each new amino acid added to the chain tugs or pushes at the others added before it. Each amino acid has an amino group ($-\text{NH}_3^+$) and a carboxyl group ($-\text{COOH}$). Peptide bonds link the carboxyl group of one amino acid to the amino group of the next amino acid. On one end of a protein, therefore, there is a free amino group called the N-terminus, and on the other end is a free carboxyl group, called the C-terminus.

The process of determining a protein's order of amino acids is called protein sequencing. A protein's sequence can easily be deduced from its gene sequence, since the order of bases on a DNA strand specifies the order in which the amino acids are linked together during translation. The chemistry involved in DNA sequencing is less complex than that which is involved in determining the order of each amino acid in an amino acid chain. There are two primary reasons why effort would be put into sequencing a protein. The first is to provide the information needed to design a synthetic DNA probe that can be used to locate the gene that codes for the protein. The second is to prove that a protein that has been isolated or manufactured in the laboratory is what it is believed to be.

Sequencing Techniques

The most widely used technique for sequencing proteins is the Edman degradation, a procedure developed by Pehr Edman in the 1950s. The reaction steps used for this method have since been completely automated by machine. The procedure uses special reagents under alternating basic and acidic conditions to remove one amino acid at a time from the protein's N-terminus. As each amino acid is released during each cycle of degradation, it is identified by chromatography, a separation technique that relies on an amino acid's unique size and electrical charge to distinguish it from the other nineteen amino acids.

In many automated approaches, high-performance liquid chromatography (HPLC) is used to tell which amino acid has been released; the amount of time it takes to travel through an HPLC column is unique to each amino acid. Up to fifty amino acids from the N-terminus can be identified using Edman degradation. If a scientist is trying to identify a previously sequenced protein, usually only the first fifteen to twenty amino acids of the purified

